

ROBINSON VIDVA

Biomedical Data Scientist

Washington, DC area

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SUMMARY

Biomedical data scientist with more than 16 years of experience in multi-omics analysis, machine learning, and data analytics pipelines for computational oncology, immuno-oncology, drug discovery, and pediatric neurodevelopment. Integrates mutational, gene-expression, imaging, and clinical data to identify biomarkers of treatment response. Founded and ran a computational biology consultancy; co-inventor on three granted patents and three published applications; released five tools as open source, including a laboratory platform in use at three laboratories; led teams of 10 or more scientists. Author of 7 journal articles and 2 preprints with 150+ citations. Uses large language models (LLMs) and agentic coding tools routinely in analysis and software development.

TECHNICAL SKILLS

Bioinformatics and multi-omics: bulk RNA-seq, metabolomics, genomics; next-generation sequencing (NGS) analysis; variant analysis (SNVs, CNVs); pathway and cell-cell communication analysis; gene-set enrichment; clinical data integration (MRI, ultrasound, neurodevelopmental scores)

Modeling and machine learning: mechanistic network, ODE, and stochastic pathway models; scikit-learn, TensorFlow, Keras, PyTorch; generalized linear, mixed-effects, segmented, and survival models

Programming and infrastructure: Python, R, SQL, MATLAB, C/C++, Unix shell, JavaScript/TypeScript, PHP; Git, CI/CD, Docker and Singularity, Snakemake, LSF, AWS; database design; full-stack web development

Behavioral phenotyping: DeepLabCut, B-SOiD, BONSAI, OpenCV, MediaPipe; three-chamber sociability and Erasmus Ladder analysis

AI-assisted research: large language models (LLMs) and agentic coding tools for analysis code, literature synthesis, and reproducibility checks

Clinical data and compliance: clinical and electronic health record (EHR) data management; removal of identifiers; research under approved Institutional Review Board (IRB) protocols; HIPAA and protected health information (PHI) handling; data-use agreements

EXPERIENCE

Research Technician (Bioinformatics)

Children's National Hospital, Washington, DC | 03/2025 - Present

- Kratimenos Lab, Sheikh Zayed Institute for Pediatric Surgical Innovation. Curate and quality-control de-identified clinical research datasets integrating electronic health record (EHR), imaging, laboratory, metabolomic, and transcriptomic data for two preterm-infant studies (178 and 181 infants).
- Perform statistical analyses relating clinical exposures to brain development using linear, mixed-effects, segmented, logistic, and modified Poisson models, with stratification and sensitivity analyses.
- Build and maintain reproducible RNA-seq pipelines for mouse and human cerebellar studies of prematurity insults, necrotizing enterocolitis, and sudden infant death syndrome; perform differential expression, pathway, and Gene Ontology (GO) enrichment analyses.
- Quantify rodent behavior with DeepLabCut and B-SOiD in a mouse model of prematurity insults and its early-rehabilitation extension; develop statistical and machine-learning models on Erasmus Ladder data from 160 preterm mice.
- Develop reproducible figure code, verify manuscripts against deposited data, and prepare sequencing-data repository submissions; contributed cancer-immunology content and figures to a co-authored review (Frontiers in Immunology, 2025).

Founder and Lead Computational Biologist

Digirobi Solutions, Bengaluru, India | 09/2019 - 06/2022

- Founded and led an independent computational biology consultancy, delivering commercial and academic engagements in cancer therapeutics and research software.
- Built cancer genomics, drug, and nutrition databases for a nutrigenomics platform, and co-invented the resulting nutraceutical combination therapy for cancer, granted in the United States and Europe.
- Built a mechanistic model of Src kinase signaling in neonatal hypoxic neuronal injury (MATLAB SimBiology), which resulted in a peer-reviewed paper (2022); designed and built MyVivarium, an open-source colony-management platform now used in three laboratories (published 2025).

Senior Lead Scientist

Cellworks Life, Bengaluru, India | 07/2007 - 09/2019

- Promoted through Associate Biomodeling Scientist (2007), Associate Lead Scientist (2008), Lead Scientist (2014), and Senior Lead Scientist (2017); mentored more than 25 junior scientists and delivered more than 25 internal training sessions.

- Led a team of more than 10 scientists developing computational precision-medicine platforms for oncology, and set the standards for model reproducibility and validation.
- Using TCGA, MSK-IMPACT, and proprietary genomic, transcriptomic, and pharmacological data, designed and validated patient-specific drug-response models across solid and hematologic cancers; in the Beat AML study, predicted sensitivity or resistance to a BET inhibitor agreed with the ex vivo IC50 classification for 93 of 100 acute myeloid leukemia (AML) patients (Leukemia Research, 2019).
- Built the cell line resource (more than 500 lines with IC50 profiles), a knowledge base of about 200 simulatable drug models, and more than 15 curated clinical data collections used to train, validate, and benchmark the models.
- Co-invented the PTEN/TP53-targeted cancer combination therapy (US 10,098,880 B2), drafting the claims and handling prosecution through examiner objections; contributed to two further patent applications.
- Built and validated an eight-cell-type systems biology model of rheumatoid arthritis against in vivo data and identified two synergistic repurposed drug combinations, one of which reduced disease progression in a murine arthritis model.

SELECTED TECHNICAL PROJECTS

- **MyVivarium:** cloud platform for lab-animal colony management with IoT sensing (PHP, MySQL, JavaScript); CSBJ 2025; in use at three laboratories; version 2 under staged deployment at the University of Dayton. github.com/myvivarium/MyVivarium
- **enrichlite:** in-browser gene-set over-representation analysis (GO, Hallmark, Reactome), human and mouse. robinsonvidva.com/enrichlite
- **drugtargets:** drug-target exploration and repurposing tool on Open Targets and openFDA. drugtargets.robinsonvidva.com
- **DrugInteract:** client-side FDA drug-interaction reference on openFDA and RxNorm. robinsonvidva.com/DrugInteract
- **neokine:** client-side MediaPipe pose visualization with joint-angle, velocity, and asymmetry plots. robinsonvidva.com/neokine

EDUCATION

M.S., Data Analytics, Western Governors University, Salt Lake City, UT

07/2024 - 01/2025

M.B.A., Project Management, Sikkim Manipal University, India

11/2012 - 02/2020

B.Tech. (Bachelor of Technology), Biotechnology, Bharathidasan University, India; First Class with Distinction

07/2003 - 04/2007

PATENTS

- EP 3,861,992 B1 (2023) and US 11,478,444 B2 (2022), granted: plant-derived supplements combined with antineoplastic compounds for hematological malignancies.
- US 10,098,880 B2 (2018), granted: nelfinavir, metformin, and rosuvastatin combination for PTEN/TP53-aberrant cancers.
- US 2015/0302167 A1, US 2015/0098993 A1, US 2014/0127295 A1, published applications: therapeutic-development method; compositions for inflammatory disease.

SELECTED PUBLICATIONS

- Vidva R, et al. MyVivarium: a cloud-based lab animal colony management application with real-time ambient sensing. *Comput Struct Biotechnol J.* 2025;27:612-23.
- Sanidas G, et al. (incl. Vidva R). Prematurity insults remodel cerebellar development and behavior; Intrinsic gestational timing governs human cerebellar development after preterm birth. *bioRxiv preprints*, 2025, revised 2026.
- Drusbosky LM, Vidva R, et al. Predicting response to BET inhibitors using computational modeling: a BEAT AML project study. *Leuk Res.* 2019;77:42-50.

Full list of 7 papers, 2 preprints, and 14 abstracts: orcid.org/0000-0002-2604-5448

Certifications: JHU Cancer Biology (2018); UCSD Drug Development Product Management (2017); UCSF Genomic and Precision Medicine (2017); JHU Clinical Trials (2019); Imperial College London immunology series (2023); Rice immunology (2020, 2021); WIPO IP for Biotechnology (2012). ISCB member since 2024.

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